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# Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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Authors: Matthijs Smakman, Ana Tanevska

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**Project Coordinator:** Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

**Project Members:**

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

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# Introduction

## 1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and hu- mans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterise and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterisation and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

# First Ethical Report

*Report by dr. Matthijs Smakman, independent ethics advisor on ethical aspects and potential misuse of research of the SYMAWARE project*

The SYMAWARE project aims to create a novel conceptual framework for situational awareness in multi-agent systems that is compatible with the internal models and specifications of robot agents and that enables the safe simultaneous operation of collaborating autonomous agents and humans. The project is mainly conceptual with a high level of technical complexity with limited involvement of human participants. The envisioned end result of the project will be a virtual environment wherein autonomous vehicles and drones can navigate.

To identify and mitigate any potential misuse of results and ethical issues the SYMAWARE project appointed an external ethical advisor dr. Matthijs Smakman, Associate Professor HU University of Applied Sciences Utrecht (author of this report).

All participating universities of the SYMAWARE project filled out a questionnaire designed to elicit potential ethical issues related to the participation of human individuals and groups in experiments, the protection of personal data, and ethical issues linked to the deployment of the novel conceptual framework this project aims for. In 2023 all six participating universities (MPI-SWS, TU/e, KTH, UU, NLR and SISW) had completed the questionnaire.

The results of this questionnaire show that only Uppsala University (UU) has worked with human individuals and has collected data from participants. Before their data collection involving human individuals they applied for ethical approval to the Swedish Ethical Review Authority (Etikprövningsmyndigheten); the application was evaluated as "not needing an ethical approval" due to the research not falling under the provisions of the Ethical Review Act of Sweden. No underaged or vulnerable participants were involved; consent forms were present and complete. See appendix A for all answers of UU to the questionnaire related to the participation of human individuals and groups in experiments and the protection of personal data.

Next to the questionnaire, a workshop has been organized with all involved groups in June 2023 in Eindhoven, the Netherlands to discuss the potential ethical implications related to the aim of the project. The workshop resulted in no (new) ethical concerns related to the results of the SYMAWARE project. The main relevant concerns related to the potential misuse of the results, such as the risks of misuse with multi-agent systems – if implemented by people who wish to harm, who might modifying risk calculation formula to select options with a maximum risk, but this is considered a very unlikely scenario.

In conclusion, I see no major ethical concerns for this phase of the SYMAWARE project. To ensure any ethical concerns arise during the project, the ethical workshop and questionnaire should have a yearly follow-up.

Sincerely,

dr. Matthijs Smakman
HU University of Applied Sciences

# *APPENDIX A - ANSWERS OF UU TO THE QUESTIONNAIRE*

**List all events in which you collected data.**

A participatory design study with users in the form of a workshop, 29/05-02/06.

- Location and date of the event
The study was conducted in Uppsala University, in the Social Robotics lab. There were four slots of 1hr30min, specifically 29/05 10:30-12:00, 30/05 10:30-12:00, 30/05 13:30-15:00, and 02/06 13:30-15:00.

- Indicate what personal data was collected/stored/processed:
We collected demographic data (age range, gender identity, driver experience), data related to the user's driving behaviour (separate questionnaire), and data related to the user's attitudes to autonomous vehicles (adapted from the NARS questionnaire on negative attitudes towards robots).
We also made audio recordings of the discussion between the participants and the experimenter.

- Data acquisition methods (cameras, microphones, behavioral data, physiological data, surveys, questionnaires, interviews, focus groups, mailing list, etc..).
Audio was recorded using an Uppsala University mobile phone. Questionnaire data was collected using three self-report questionnaires. The workshop data consisted of flip chart drawings and collage of post-its.

- Anonymisation or pseudo--anomysation method (blurring), removal of data.
The self-report questionnaires were filled using a randomized 8-character ID that participants generated themselves, their names and surnames are not saved with any of the collected data. Audio recordings are not anonymized though.

- Number of participants
In total there were 11 adult participants, split in 4 groups.

- Methods of data processing (whether those operations are automated or not)
Audio data will be manually coded and analyzed. Survey data will be manually analyzed in SPSS or another statistics software. Workshop data will be manually processed and analyzed.

- Data to be kept (and for how long), data to be destroyed (if any) and how?
Data will be stored on servers owned by Uppsala University. Collected data will be processed as long as necessary to ensure the quality of the research, and at least for the whole duration of the SymAware project.

- Data and participant consent forms storage methods (saved on secured servers, protected by password, control access etc..)
Data will be stored on servers owned by Uppsala University.

- Strategy to handle subject access request
Participants can request the audio recordings of their experimental slot. Survey data can be requested if participants retrieve it with their unique ID>

- Strategy to report data breaches
Incidents during the processing (data loses, intrusion, unwanted deletion etc.) will be reported to the data protection officer at Uppsala University (dataskyddsombud@uu.se).

**Did you apply for ethical approval for research involving human participants or personal data?(if Yes, please elaborate)**

Yes, we applied to the Swedish Ethical Review Authority (Etikprövningsmyndigheten). The application was evaluated as "not needing an ethical approval" due to the research not falling under the provisions of the Ethical Review Act of Sweden.

**Describe the criteria you used for identification of participants (in order to highlight anycritical issues related to selection by gender, age, political opinion or religious, sexual orientation, etc). Describe inclusion and exclusion criteria.**

Inclusion criteria was age of 18 years or above. No other criteria.

**Describe the procedures you used for the recruitment of participants**

Participants were recruited via advertisements on the Uppsala University notice boards. Additionally, advertisement was done via email in courses taught at the department of information technology. Participants then signed up via an online booking system (Doodle).

**Describe the procedures you used to inform participants about the purpose of the research.**

Participants were presented with an Information sheet that they could keep after the experiment. Any further questions were answered on the spot by the experimenter.

No minor participants.

**Describe the procedures you used to inform participants about the use of collected information and the storage of data.**

Participants were informed of this via the information sheet and the consent form.

**Describe the procedures you used to inform participants, including children, that they are free to withdraw from participation at any point.**

Participants were informed of this both verbally and via the information sheet.